

Mohammad Orabe

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in mohammad-orabe

I am a biomedical data scientist and software engineer with 4+ years of experience in deep learning, human-compute interaction, computer vision, and multimodal signal analysis. I develop software systems that tackle challenges at the intersection of these domains. My portfolio is available on my website.

EDUCATION

M. Sc. Computer Science – Technische Universität (TU Berlin) Oct 2022 – Jan 2026

- Focus: Machine Learning, Data Science, Intelligent Systems, and Human-Computer Interaction.
- Thesis: Deep Learning-Based Decoding of Invasive Brain Signals for Adaptive Deep Brain Stimulation

B. Sc. Computer Science – Technische Universität (TU Berlin) Oct 2018 – Sep 2021

- Focus: Software Engineering, Distributed Systems, Cloud Computing, Computer Architecture
- Thesis: Pose Estimation and Multi-Animal Tracking with Deep Learning and Computer Vision.

EXPERIENCE

Applied ML Researcher (Part-Time) – Germany's National Metrology Institute Feb 2025 – Present

- Developed a framework for uncertainty quantification and calibration in M/EEG inverse ML modeling.
- Integrated data-driven uncertainty estimation into Bayesian methods for neuroscience applications.
- Built a real-time intelligent system (aDBS) for decoding impaired gait from invasive brain signals.

Signal Analysis Engineer (Part-Time) – Quality in AI Labs | Charité (Berlin) Jul 2023 – Dec 2024

- Implemented advanced signal processing methods for electrophysiological data.
- Extended multivariate connectivity analysis in MNE, a widely used open-source neuroimaging toolkit.

Data Science Research Assistant – Winter Lab | HU (Berlin) Jul 2021 – Jun 2023

- Developed ML pipelines and statistical toolboxes across five projects in animal tracking environments.
- Applied ML and computer vision to biomedical sensor data for animal detection and pose estimation.

IT Administration Assistant – Institute for International Communication (Berlin) Feb 2017 – Jul 2020

- Maintained OS systems, supported internal network and user infrastructure.
- Diagnosed and resolved hardware/software issues; optimized administrative workflows.

HONORS & AWARDS

ML Research Stipend – BIFOLD | Intelligent Biomedical Sensing (Berlin) Sep 2025 – Mar 2026

- Awarded a competitive €4,200 stipend to conduct research on deep generative models for reconstructing high-resolution brain signals from sparse wearable fNIRS recordings.

MENTORSHIPS

Founder & Lead Instructor – MathCodeLab (Berlin) Aug 2023 – Present

- Founded a tech education academy offering in-person/online courses in coding, math & data science.
- Designed and taught 15+ intensive 12-week courses to undergraduate students, building a growing learner cohort of over 250 students.

Teaching Assistant – Chair Uncertainty, Inverse Modeling & ML (TU Berlin) Oct 2024 – Apr 2025

- Assisted in a Master's-level seminar focused on advanced machine learning for biomedical data science. Supported students with project supervision, course logistics, and technical Q&A.

INTERNSHIPS

IoT Middleware Developer – D-AI Lab | GT-ARC (Berlin)

Jul 2020 – Oct 2020

- Implemented components for a holistic middleware to enable communication across IoT devices.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, Java, MATLAB, C/C++
- **ML Frameworks:** TensorFlow, PyTorch, Keras, scikit-learn, NumPy, Pandas, SciPy, OpenCV
- **Development:** Git, Bash, Docker, CI/CD, SLURM, MLOps
- **ML Domains:** BCI, Biomedical Signal, Time-Series Analysis, Computer Vision, Multimodal HCI

OPEN-SOURCE TOOLBOXES (LEAD DEVELOPER)

CaliBrain: Model Calibration – Germany's National Metrology Institute (PTB)

Developed a benchmarking framework to evaluate and calibrate uncertainty in data-driven inverse problems in electromagnetic source imaging (Python).

GaitMod: Deep Learning for Signal Decoding – PTB | Charité

Built a toolkit for decoding electrophysiological brain data with deep learning models (Python).

GroupMouseTrack: Multi-Modal Tracking – Winter Lab

Developed a deep learning system integrating RFID data with computer vision to enable robust pose estimation and identity-preserving multi-object tracking over time (Python).

ColonyRack: Behavioral Data Processing – Winter Lab

Created a toolbox for processing and analyzing large-scale time-series data in behavioral studies to enable efficient extraction of activity patterns and social interaction metrics (R).

ML UNIVERSITY PROJECTS (SELECTED)

Medical Image Segmentation – University Project

Built segmentation models (UNet, SegResNet) for identifying brain tumors in MRI scans.

Multi-Modal Generative Learning – University Project

Built a multimodal variational autoencoder (VAE) to enable format translation and joint inference.

Neural Simulation Framework (neurolib) – University Project

Extended an open-source library (neurolib) for simulating brain activity based on anatomical models.

PUBLICATIONS & TALKS

Orabe, M., Haufe, S., Isaias, I. Towards Closed-Loop Deep Brain Stimulation: A Deep Learning Approach for Real-Time Gait Classification in Parkinson's Disease, (In Preparation, early 2026).

Orabe, M., Haufe, S., Huseynov, I. Calibrain: A Framework for Calibrated Uncertainty Quantification in M/EEG Source Imaging, (In Preparation, early 2026).

Huseynov, I., **Orabe, M.**, Hashemi, A., Nagarajan, S., Haufe, S., Establishing a framework to measure uncertainty calibration in M/EEG source imaging, (In Preparation, early 2026).

Huseynov, I., **Orabe, M.**, Hashemi, A., Nagarajan, S., Haufe, S. (2025). A Framework for Uncertainty Calibration in M/EEG Source Imaging. *Brain Quantum Imaging, Berlin, Germany. (Poster)*

Binns, T.S., **Orabe, M.**, Nguyen, T.D., Köhler, R.M., Pellegrini, F., & Haufe, S. (2024). Multivariate connectivity methods in the MNE-Python toolbox. *Neural Traces, Berlin, Germany. (Software Poster)*